

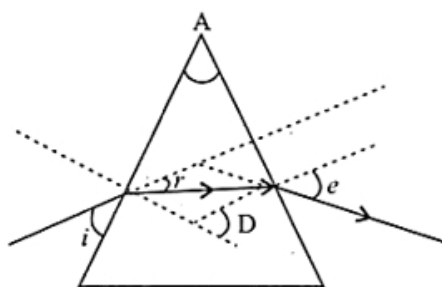
HUMAN EYE -02

Class 10 - Science

Section A

1. In the following diagram the correctly marked angles are:

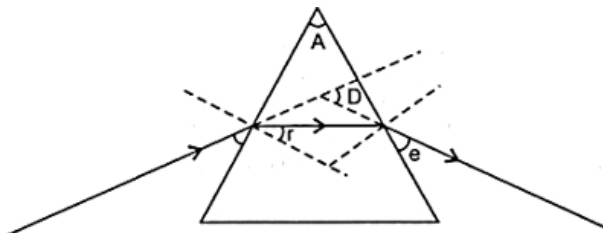
[1]



- a) $\angle A$, $\angle i$ and $\angle e$
b) $\angle A$, $\angle r$ and $\angle D$
c) $\angle i$, $\angle A$ and $\angle D$
d) $\angle A$ and $\angle e$

2. In the following ray diagram the correctly marked angles are:

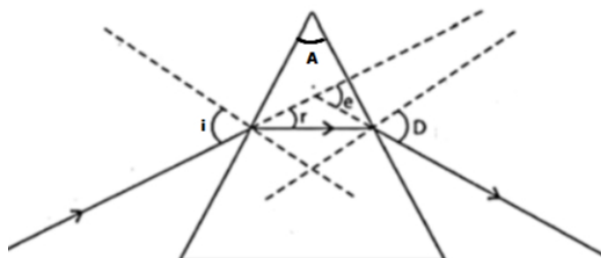
[1]



- a) $\angle i$ and $\angle e$
b) $\angle r$, $\angle A$ and $\angle D$
c) $\angle i$, $\angle e$ and $\angle D$
d) $\angle A$ and $\angle D$

3. Study the following figure in which a student has marked the angle of incidence ($\angle i$), angle of refraction ($\angle r$), angle of emergence ($\angle e$), angle of prism ($\angle A$) and the angle of deviation ($\angle D$). The correctly marked angles are:

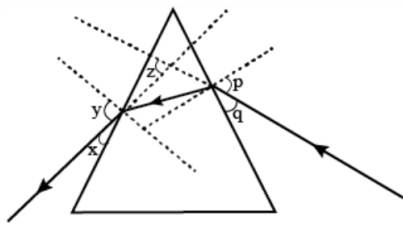
[1]



- a) $\angle A$, $\angle i$ and $\angle r$
b) $\angle A$, $\angle i$, $\angle r$ and $\angle D$
c) $\angle A$, $\angle i$ and $\angle D$
d) $\angle A$ and $\angle i$

4. Study the following ray diagram

[1]



In this diagram, the angle of incidence, the angle of emergence and the angle of deviation respectively have been represented by:

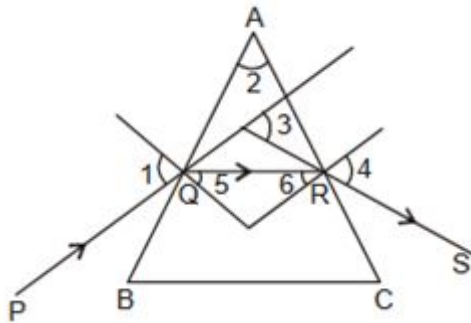
a) p, y, z

b) y, p, z

c) x, q, z

d) p, z, y

5. What is the relation between angles 1, 2, 3 and 4? [1]



a) $\angle 2 + \angle 3 = \angle 1 + \angle 4$

b) $\angle 1 + \angle 2 = \angle 3 + \angle 4$

c) $\angle 2 + \angle 3 + \angle 4 = \angle 1 + \angle 4$

d) $\angle 1 + \angle 3 = \angle 2 + \angle 4$

Section B

6. Match the following: [2]

(a) Rod cells	(i) Image formation
(b) Cone cells	(ii) Control amount of light
(c) Pupil	(iii) Colors
(d) Retina	(iv) Light

7. Match the following: [2]

(a) Pupil	(i) infinity
(b) Iris	(ii) 25 cm
(c) Near point	(iii) color of eye
(d) Far point	(iv) control amount of light

8. Match the following: [2]

(a) Red color	(i) Atmospheric refraction
(b) Violet Color	(ii) Dispersion
(c) Visible spectrum	(iii) Deviates maximum
(d) Twinkling of stars	(iv) Deviates the least

9. Match the following: [2]

(a) Ciliary muscles	(i) 25 cm
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(b) Retina	(ii) Hole in the iris
(c) Pupil	(iii) Image is formed
(d) Near Point	(iv) holds eye lens

10. Match the following: [2]

(a) The spectrum of white light	(i) 25 cm
(b) Delayed sunset	(ii) Scattering of light
(c) Tyndall effect	(iii) Atmospheric refraction
(d) Near point	(iv) Dispersion

11. Match the following: [2]

(a) Sky looks blue	(i) Long-sightedness
(b) Advanced sunrise	(ii) Short-sightedness
(c) Myopia	(iii) Atmospheric refraction
(d) Hypermetropia	(iv) Scattering of light

12. Match the following: [2]

(a) Myopia	(i) Rainbow
(b) Hypermetropia	(ii) scattering of light
(c) Tyndall effect	(iii) low converging power of the lens
(d) Dispersion	(iv) high converging power of the lens

13. Match the following: [2]

(a) Iris	(i) Eye ball being too long
(b) Rod cell	(ii) Color
(c) Cone cell	(iii) Light
(d) Myopia	(iv) Color of eye

14. Match the following: [2]

(a) Eyeball too short	(i) Atmospheric refraction
(b) Eyeball too long	(ii) hole in iris
(c) Pupil	(iii) Myopia
(d) Twinkling of stars	(iv) Hypermetropia

15. Match the following: [2]

(a) Rainbow	(i) deviates least
(b) Twinkling of stars	(ii) Dispersion of light
(c) VIBGYOR	(iii) Atmospheric refraction
(d) Red color	(iv) Dispersion of light

16. Match the following: [2]

(a) Presbyopia	(i) controls the amount of light
(b) Myopia	(ii) long-sightedness
(c) Hypermetropia	(iii) short-sightedness
(d) Pupil	(iv) defect of eye

17. Match the following: [2]

(a) Iris	(i) Long-sightedness
(b) Pupil	(ii) Short-sightedness
(c) Myopia	(iii) Light enters through it
(d) Hypermetropia	(iv) Color of eye

18. Match the following: [2]

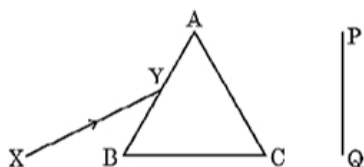
(a) Blue color	(i) Short-sightedness
(b) Red colour	(ii) Atmospheric refraction
(c) Twinkling of stars	(iii) Scattered least
(d) Myopia	(iv) Scattered most

19. Match the following: [2]

(a) Dispersion	(i) long-sightedness
(b) Refraction	(ii) short-sightedness
(c) Myopia	(iii) bending of light
(d) Hypermetropia	(iv) seven colors

Section C

20. If Ram, a myopic person uses spectacles of power -0.5 D , then what will be the distance of far point of his eye? [1]
21. What is scattering of light? Why does the clear sky appear blue? [1]
22. While doing the experiment to trace the path of rays of light through a glass prism what should be done in the beginning of experiment and why? [1]
23. Name the principle on which a prism form the image of an object? [1]
24. What happens to the lens and the ciliary muscles when you are looking at nearby objects? [1]
25. A narrow beam XY of white light is passing through a glass prism ABC as shown in the diagram: [1]

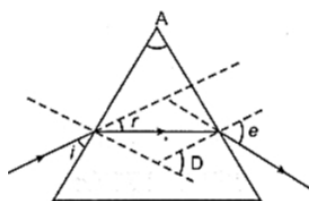


Trace it on your answer sheet and show the path of the emergent beam as observed on the screen PQ.

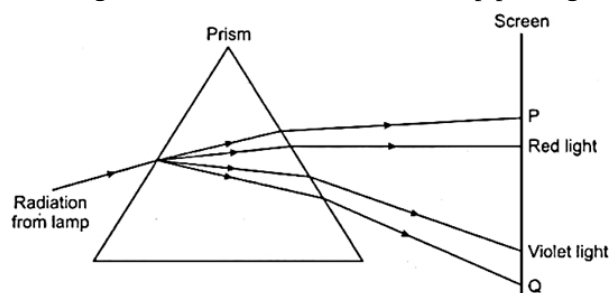
Name the phenomenon observed and state its cause.

26. What do you mean by a rainbow? [1]
27. Define power of accommodation? [1]
28. Which phenomenon is responsible for increasing the apparent length of the day by 4 min? [1]

29. Where is the image formed in an eye suffering from long sightedness? [1]
30. What is the relation between intensity of scattered light and its amplitude? [1]
31. What would have been the colour of the sky, if the Earth had no atmosphere? [1]
32. Why are danger light signals red in color? [1]
33. In which direction a ray of light bends while emerging out of a prism? [1]
34. When sunlight enters a dark smoke filled room, its path becomes visible. Name the phenomenon responsible for this. [1]
35. It is observed that the power of an eye to see nearby objects as well as far off objects diminishes with age. [1]
- Give reason for the above statement.
 - Name the defect that is likely to arise in the eyes in such a condition.
 - Draw a labelled ray diagram to show the type of corrective lens used for restoring the vision of such an eye.
36. Which component of white light is least scattered by fog or smoke? [1]
37. A person needs a lens of power -4.5 D for the correction of her vision. What is the focal length of the corrective lens? [1]
38. Variable focal length of the eye lens is responsible for which property of eye? [1]
39. In the following ray diagram which angles are properly labelled and what do we call these angles? [1]

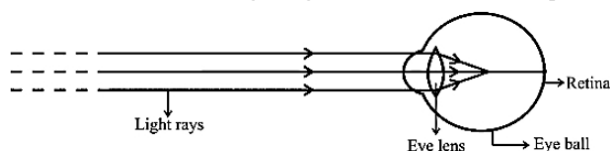


40. The diagram shows radiations from a lamp passing through a prism. [1]



Which type of radiations are found at P and Q?

41. Bees are able to see ultraviolet light. Comment. [1]
42. The colour of clear sky from the earth appears blue but from the space it appears black. Why? [1]
43. Name the part of the eye: [1]
- that controls the amount of light entering into the eye.
 - that has real, inverted image of the object formed on it.
44. Name the part of our eyes that helps us to focus near and distant objects in quick succession. [1]
45. Which part of the human eye conveys the electrical signals generated by the light sensitive cells of the retina to the brain? [1]
46. Though a hot iron emits radiations, yet it is not visible in the dark, why? [1]
47. Observe the following diagram and answer the questions following it: [1]



- Identify the defect of vision shown.

- ii. List its two causes.
- iii. Name the type of lens used for the correction of this defect.
48. What is the function of the liquid filled in the space between the eye lens and retina? [1]
49. What is the function of the crystalline lens of the eye? [1]
50. A person is advised to wear spectacles with convex lenses. What types of defect of vision is he suffering from? [1]
51. On which factor does the color of the scattered light depends? [1]
52. How can you identify the type of defect of vision a person is suffering from by physically touching his spectacles? [1]
53. A man is wearing glasses of focal length +1m, what can be defect in the eye? [1]
54. Why is the colour of the clear sky blue? [1]
55. Name the factor on which the colour of light depends? [1]
56. Where do we see the following lenses in a bifocal spectacle? [1]
- i. concave lens
- ii. convex lens
57. Where is the image formed in an eye suffering from near-sightedness? [1]
58. A person needs a lens of power -4.5 D for the correction of her vision. What is the nature of the corrective lens? [1]
59. When and where does a rainbow appear in the sky? Draw a labelled ray diagram to show its formation. [1]
60. Mala can see the objects lying between 25 cm and 100 cm from her eye. Her vision can be corrected by using lens of power -0.1 D . Is the statement true or false? [1]
61. A person needs a lens of power -4.5 D for the correction of her vision. What kind of defect in vision is she suffering from? [1]
62. If the far point of Raju, a myopic person is 4 m, then how would you calculate the power of the lens he requires to look at the stars? [1]
63. A piece of cloth appears red in sunlight. How will it appears when held in the blue portion of a spectrum? [1]
64. Which part of the human eye provides most of the refraction for the light rays entering the eye? [1]
65. Why do we observe difference in colours of the Sun during sunrise, sunset and noon? [3]
66. When one enters a less lighted room from a place of intense light, he is not able to see anything for sometime, but after sometime the things become somewhat visible. Explain how this happens? [3]
67. 1. Write the function of each of the following parts of human eye : cornea, iris, crystalline lens, ciliary muscles. [3]
2. Millions of people of the developing countries of world are suffering from corneal blindness. These people can be cured by replacing the defective cornea with the cornea of a donated eye.
- A charitable society of your city has organised a campaign in your neighbourhood in order to create awareness about this fact.
- If you are asked to participate in this mission how would you contribute in this noble cause?
1. State the objective of organising such campaigns.
2. List two arguments which you would give to motivate the people to donate their eyes after death.
3. List two values which are developed in the persons who actively participate and contribute in such programme.
68. i. State two main causes of a person developing near-sightedness. With the help of a ray diagram, suggest how he can be helped to overcome his disability? [3]
- ii. The far point of myopic person is 100 cm in front of the eye. Calculate the focal length and power of a lens required to enable him to see distant objects clearly.

69. A camera in many ways is similar to the human eye, still, there are some basic differences in image formation between the two. Explain. [3]

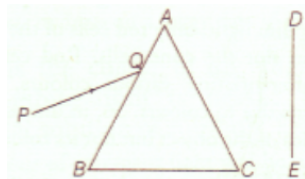
70. How can changes of size of eyeball be one of the reason for [3]
i. myopic and
ii. hypermetropic eye?

Compare the size of eyeball with that of a normal eye in each case. How does this changes of size affect the position of image in each case?

71. A student sitting at the back of the classroom cannot read clearly the letters written on the backboard. What advice will a doctor give to her? [3]

72. Why do different rays deviate differently in the prism? [3]

73. A narrow beam PQ of white light is passing through a glass prism ABC as shown in the diagram. [3]



Trace it on your answer sheet and show the path of the emergent beam as observed on the screen DE.

- Write the name and cause of the phenomenon observed.
 - Where else in nature is this phenomenon observed?
 - Based on this observation, state the conclusion which can be draw about the constituents of white light.
74. What are the common defects of vision that can be corrected by the use of suitable eyeglasses or spectacles? [3]